End-to-End Solution for Braille-To-Chinese

Yantian Zheng
Zhejiang Provincial Key Laboratory of Service Robot
School of Software Technology
Zhejiang University
Ningbo, China
zhengyantian@zju.edu.cn

Hangdi Xing
Zhejiang Provincial Key Laboratory of Service Robot
College of Computer Science
Zhejiang University
Hangzhou, China
xinghd@zju.edu.cn

Zhi Yu
Zhejiang Provincial Key Laboratory of Service Robot
School of Software Technology
Zhejiang University
Ningbo, China
yuzhirenzhe@zju.edu.cn

Xinhua Zhang
Zhejiang School for the Blind
Hangzhou, China
zxhzjmx@126.com

Feiyu Gao
DAMO Academy
Alibaba Group
Hangzhou, China
feiyu.gfy@alibaba-inc.com

Jiajun Bu
Zhejiang Provincial Key Laboratory of Service Robot
College of Computer Science
Zhejiang University
Hangzhou, China
bjj@zju.edu.cn

2023 IEEE International Conference on Sensors, Electronics and Computer Engineering (ICSECE) | 979-8-3503-1373-4/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICSECE58870.2023.10263505

***Abstract*—In the field of education, Teachers of Students with Visual Impairments need to read and review student's Braille homework. Reading Braille consumes a lot of teachers' time and energy. We propose an end-to-end solution to translate Braille into Chinese. Our approach uses a deep learning method to detect and recognize Braille in images and translate them into Chinese and mathematical formulas. Finally, we reconstruct the translation results into the style of the input Braille image. Our approach is able to adapt to more complex Braille images and make the translation results easier to read and understand, significantly improving the speed of reading Braille for non-disabled.**

Keywords—Braille, Braille Education, Object Detection, Chinese Braille, Braille Translate

I. INTRODUCTION

According to a report by the World Health Organization, there are currently 284 million people in the world who are visually impaired. Braille education is an important part of learning Braille. Reading and writing Braille is one of the tools used by visually impaired people to access information and communicate with others who use their tactile nerves [10].

However, there are some difficulties with Braille education in practice. Most of the Teachers of Students with Visual Impairments (TSVI) are non-disabled. Reading Braille is difficult for them. They read Braille slower than their native language and they take a great deal of time to read students' Braille. The huge cost of learning Braille also prevents non-disabled from getting involved in Braille education. Most parents of visually impaired children also do not understand Braille. They are unable to provide adequate assistance in the development of their children. These problems have hindered the development of Braille education.

Much of the work in the past has been aimed at helping non-disabled join Braille education more easily. Most of the past work has been devoted to the interconversion of different languages to Braille. Some work has translated books into Braille to make reading materials more accessible to visually impaired. For instance, Taekeun Park [9] proposed a method for automatically translating scanned images of printed books into electronic Braille books, which could reduce the time and cost required to produce Braille books. Other work has translated input Braille text into different languages. For instance, Minghu Jiang [5] proposed a method to convert Braille into Chinese. They used the N-Best search algorithm to translate Braille into Chinese characters.

Past work on Braille translation has not always been feasible in Braille education. Most of them only support the translation of Braille text, ignoring other types of Braille that may be present in a Braille document. For instance, A Chinese Braille examination paper may contain not only Chinese but also tables and pictures. Once the input Braille contains meaningless Braille text used to describe the structure, it can significantly reduce their translation effectiveness.

We solve this problem for Braille translation with a new process that eliminates the user input step and recognizes the Braille image scanned by the scanner directly. Then we use deep learning approaches to analyze the structure of the Braille image, dividing it into sections according to the type of Braille. For each type of Braille, we design different processing methods to translate them.

We propose an end-to-end Braille-to-Chinese translation system, named "Read Light". We use deep learning techniques to recognize Braille in Braille Document images and abstract the layout information of documents from the recognition results. Then we translate the recognized Braille into Chinese or mathematical formulas according to the rules of Braille dot transcription (including Chinese and mathematical Braille) of the National Standard for Braille. Our method was able to translate Braille image containing both Chinese Braille and mathematical Braille, while experiments showed that this also improved the accuracy of translation in some cases. In order to verify the usability of the system, five subjects were interviewed and the system was evaluated in detail and in depth. The results showed that our system significantly improved the speed of reading Braille for non-disabled, thus improving the effectiveness of Teachers of Students with Visual Impairments.

Authorized licensed use limited to: University of New South Wales. Downloaded on December 02,2025 at 14:30:55 UTC from IEEE Xplore. Restrictions apply.

II. Related Work

A. Braille Detection and Recognition

Past approaches have mainly used machine learning methods to detect Braille dots. Shanjun Zhang[13] used fast dynamic thresholding and image histogram to detect Braille dots. Then they grouped obtained Braille dots according to the horizontal coordinates (x-coordinates) and assigned Braille dots into 3x2 grid by calculating average distance between groups. Renqiang Li [6] used Haar and Adaboost method to detect Braille dots in images and make post-processing processes similar to the above. Their method got F1=97 at the Braille dot level. These methods are very sensitive to the setting of thresholds and processing of combining Braille dots into Braille characters requires that images are not seriously skewed.

With the development of deep learning, neural network-based methods surpass machine learning methods in various fields. For the first time, Ilya G. Ovodov [8] was able to directly detect character-level Braille using a deep learning algorithm. The method got F1=99.8 at the Braille character level and F1=99.9 at the Braille dot level based on deep learning methods. Neural network-based methods are not only more accurate than traditional methods, but also more robust. The method can accept little deformation of images.

B. Braille Translation

Letters and Braille can be mapped one to one the other because the number of letters in most languages is less than the number that Braille can represent. K. Sasindu Anuradha [1] mapped Braille characters to English letters and Sinhala one by one, and then rearranges them into readable text. The translation accuracy in the general case is 85.30%. Ju-Xiao Zhang [12] created a Chinese Braille dictionary and used a binary language model lexicon to solve the one-to-many problem in Pinyin to Chinese characters mapping. Minghu Jiang[5] used N-Best method to find the maximum possible N Chinese character sequences. The best sequence is selected as the result by calculating the scores of these sequences in word graph. The method has 94.38% translation accuracy.

III. Method

A. Overview

Our method receives images scanned by the scanner and uses deep learning techniques to detect and recognize Braille dots in the image. Elements in Braille image are then detected and classified. For each Braille point, we assign it to one type: Chinese, mathematical or table. We design different processing methods for each of the different types. Finally, the translation results are reconstructed according to the position of the Braille dots to produce easily understood visual results. Fig. 1 illustrates the architecture and procedure of our system.

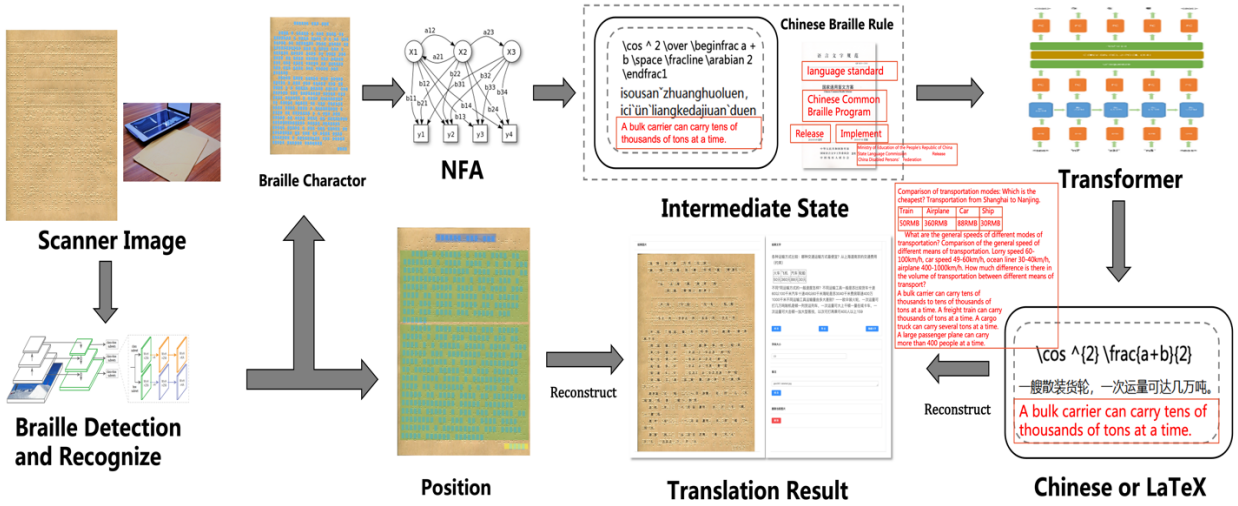


Fig. 1. Framework of End-to-End Braille Translation System(Red words are translations of Chinese words)

B. Braille Character Detection and Recognition

This stage focuses on detecting and recognizing the position information and dot information of Braille characters. We follow the approach of Ilya G Ovodov [8] using RetinaNet [7] object detection method to detect the position and classes of Braille characters in parallel. We consider Braille character as a rectangle. RetinaNet network directly predicts the coordinates of the four corners of the rectangle to which Braille character belongs and the category of each character. A Braille character consists of a 3x2 rectangle with the first column numbered 1, 2, 3 from top to bottom and the second column numbered 4, 5, 6. The character class is a string describing Braille dots information. Braille dot number in the string means that the position is dot and Braille dot number not in the string means that there is no dot. For example, "126" means that positions numbered 1, 2, 6 have dots and positions numbered 3, 4, 5 have no dots. To improve the accuracy of the model, we reduce the size of candidate anchors based on the a priori knowledge of Braille but do not significantly affect the speed of inference. The result of Braille recognition is the cornerstone of the whole translation system. Accuracy of detection directly and significantly affects the translation results. Because Errors will be amplified in subsequent process.

C. Table Structure Detection

Braille images may contain Braille Characters used to describe the structure of table. We use CenterNet [4] object detection method to detect tables in the input images. Its

Authorized licensed use limited to: University of New South Wales. Downloaded on December 02,2025 at 14:30:55 UTC from IEEE Xplore. Restrictions apply.

detection strategy of segmenting key points is well suited for detecting wired forms in Braille document images. These Braille characters are detected and then removed from Braille character detection and recognition results. These Braille characters are not involved in the subsequent translation process. They are only converted into HTML table elements in visual output.

D. Text Translation

The Chinese national standard of Mandarin Braille defines the conversion rules for Chinese Braille. Chinese Braille is a phonemic alpha representation. The conversion process can be divided into two steps: from Braille to Pinyin and from Pinyin to Chinese characters. We use a nondeterministic finite automaton (NFA) to accomplish the transcription of Braille sequences to intermediate states. In practice, we found a serious one-to-many confusion problem in conversion process. Braille has only 64 different dots in total, which cannot represent numbers and symbols in mathematical formulas one by one. Same Braille square has different meanings in different cases. We add a backtracking mechanism to the automaton when state transition is uncertain. When system fails to determine what Braille represents, one choice, such as the letter "a", is tried first. If the conversion fails, the backtracking mechanism is activated.

We design end-to-end encoders and decoders based on Transformer [11, 3] to transcribe pinyin into Chinese characters according to Chinese Braille rules. Model directly segments and translates consecutive pinyin sequences into Chinese character sequences.

In addition, many abbreviation rules have been designed in the Braille standard in order to simplify Braille writing. We try to recover omitted symbols during the automatic machine transcription. For omitted symbols that are difficult to recover, further attempts are made to recover them in model's encoders and decoders.

E. Math Formula Translation

Chinese mathematical Braille is a one-dimensional formula. Besides defining the braille for numbers and mathematical symbols, some auxiliary braille parties are defined in the rules to represent the structural information of the formula. We still use NFA to convert it from Braille to a one-dimensional form. It is the same as the Chinese. Then we designed a set of symbol generation rules based on the LaTeX structured algorithm to transcribe one-dimensional mathematical formulas into LaTeX formulas directly according to the rules of Chinese mathematical Braille and the rules of LaTeX formula.

F. System Implement

We have developed a visualization system to implement our approach. The system takes an image as input and detects Braille dots and Structure in input Braille image. Based on position information in structure detection result and Braille character detection and recognition result, Braille describing the structure is removed from result. Remaining Braille is fed into the NFA, which attempts to convert the Braille into either pinyin or mathematical formulas. Pinyin sequence generated by the NFA is fed into Transformer-based decoder. Mathematical formulas are transcribed into LaTeX formulas according to rules we have designed. Finally, Chinese and LaTeX sequences are output as a result for visualization. In order to make result of translation easier to read and understand. We sort translation results by rearranging them according to the order in which Braille characters are written in the input image. We use positional information obtained in Braille detection and recognition to connect translation results to Braille dots. Each word is then transcribed into an HTML(HyperText Markup Language) element in top-to-bottom order and rendered by the browser for output.

IV. EVALUATION

We used 30 scanned images from Chinese junior high school textbooks as our test data, including Chinese, mathematics， geography and political science subjects. We use the Character Error Rate (CER) metric to evaluate the translation accuracy of the system, which uses Levenshtein Distance of two sequences to evaluate the similarity of two sequences. The lower CER score indicates better translation performance.

TABLE I. CER OF TRANSLATION RESULTS

No	*Image type*	*Table Module*	*Math Module*	*CER*
1a	***Chinese***	***Without***	***Without***	***0.0677***
1b	***Chinese***	***With***	***Without***	***0.0677***
1c	***Chinese***	***Without***	***With***	***0.0677***
1d	***Chinese***	***With***	***With***	***0.0677***
2a	***Chinese+Math***	***Without***	***Without***	***0.05316***
2b	***Chinese+Math***	***With***	***Without***	***0.05316***
2c	***Chinese+Math***	***Without***	***With***	***0.04253***
2d	***Chinese+Math***	***With***	***With***	***0.04253***
3a	***Chinese+Table***	***Without***	***Without***	***1.99432***
3b	***Chinese+Table***	***With***	***Without***	***0.06051***
3c	***Chinese+Table***	***Without***	***With***	***1.99432***
3d	***Chinese+Table***	***With***	***With***	***0.06051***
4a	***Chinese+Table+Math***	***Without***	***Without***	***3.06569***
4b	***Chinese+Table+Math***	***With***	***Without***	***0.37845***
4c	***Chinese+Table+Math***	***Without***	***With***	***2.69343***
4d	***Chinese+Table+Math***	***With***	***With***	***0.25564***

TABLE I. shows average CER for different types of Braille images with and without table detection and mathematical modules. Our method improves CER when pictures contain tables and math symbols. The average CER for image translation consisting of Chinese only was 0.0677. Table detection and math detection did not affect the translation results, as shown in experiment 1a, 1b, 1c and 1d. According to the comparison between experiment 2a and 2c, adding the math detection boosts the system performance on images containing Chinese and math symbols, getting a CER of 0.04253, which was about 0.1 better than the CER with math detection removed. While the table detection module has a significant influence on the CER, as depicted in experiment 3a and 3b (or 3c and 3d). The CER without filtering tables in images containing tables is only 1.99432, which is 1.93 worse than the system equipped with table detection module. Because it requires a larger number of braille to describe the table than that within the table. These Braille also span multiple lines, splitting the correct Braille sequence.

Łukasz Bola [2] reported that persons without disabilities are able to read Braille at an average speed of 6 words per minute after 9 months study, which is far below the final reading speed of visually impaired people. This speed also increases slowly with an average improvement rate of 6-9 words per minute per year. Normally, persons without disabilities do not have so much time to train Braille reading skills. We selected 5 volunteers without disabilities and grasp of Braille to test their Braille reading speed after using our system.

Authorized licensed use limited to: University of New South Wales. Downloaded on December 02,2025 at 14:30:55 UTC from IEEE Xplore. Restrictions apply.

TABLE II. shows the speed of Braille letters for sighted people after training [2] and using our system. The reading speed with our system greatly exceeds the Braille reading speed achieved by people.

This experiment shows that our system not only allows non-disabled to read Braille without any basic Braille knowledge, but also reads Braille quite fast.

Images in the experiment are all Chinese Braille images. Although every country uses Braille, different countries have different standards for translating Braille into human languages. Therefore, our system can only translate Chinese, which is a limitation of our system.

TABLE II. SPEED OF READING BRAILLE LETTER

Months of Learning Braille	*Number of Braille Read(1min)*
0	0
1	4
6	15
9	18
0(ours)	643

Fig. 2. shows that our method facilitates people who do not know Braille to read Braille at speed. Persons read faster in the reconstructed results because they can easily distinguish the end of a paragraph without being bothered by 2 sentences that have completely different meanings.

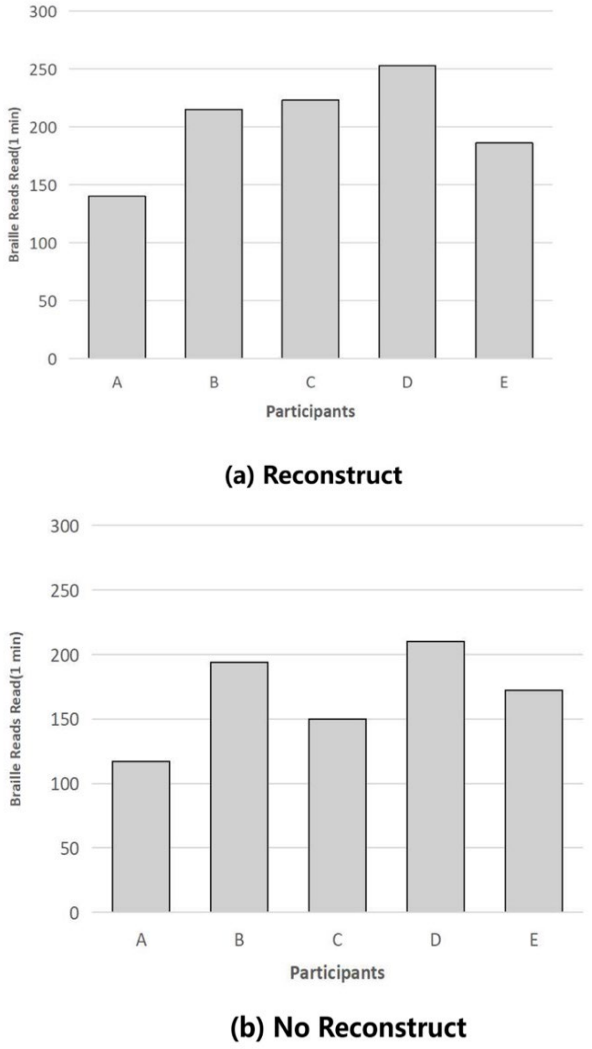


Fig. 2. Speed of Reading Braille Words

In summary, our method works on more complex Braille images and produces results that are closer to the original Braille images and human reading habits. People who do not know Braille can also read Braille quickly.

V. CONCLUSION

We have designed and implemented an end-to-end solution to translate images of scanned Braille documents. Our approach detects Braille text and document structure from the image and reconstructs outputs after translation. Our approach produces results that are much closer in semantics and structure to the original Braille documents.

We designed a visualization system to implement the process we proposed. The system receives a scanned image, performs the process of detection and translation, and finally renders translation result through browser and Latex.

Our evaluation shows that our method not only significantly improves the accuracy of translation in images containing tables and mathematics but also improves the speed of reading Braille for the average person. We are convinced that our method is useful in many areas, especially in the field of education of visually impaired students.

ACKNOWLEDGMENT

This work is supported by the National Key Research and Development Program (Grant No. 2022YFF0902003).

REFERENCES

[1] K Sasindu Anuradha and Samantha Thelijjagoda. 2020. Machine translation system to convert Sinhala and English Braille documents into voice. In 2020 International Research Conference on Smart Computing and Systems Engineering (SCSE). IEEE, 7–16.

[2] Łukasz Bola, Katarzyna Siuda-Krzywicka, Małgorzata Paplińska, Ewa Sumera, Paweł Hańczur, and Marcin Szwed. 2016. Braille in the sighted: Teaching tactile reading to sighted adults. PloS one 11, 5 (2016), e0155394.

[3] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805 (2018).

[4] Kaiwen Duan, Song Bai, Lingxi Xie, Honggang Qi, Qingming Huang, and Qi Tian. 2019. Centernet: Keypoint triplets for object detection. In Proceedings of the IEEE/CVF international conference on computer vision. 6569–6578.

[5] Minghu Jiang, Xiaoyan Zhu, Georges Gielen, Elliott Drábek, Ying Xia, Gang Tan, and Ta Bao. 2002. Braille to print translations for Chinese. Information and software Technology 44, 2 (2002), 91–100.

[6] Renqiang Li, Hong Liu, Xiangdong Wang, and Yueliang Qian. 2018. DSBI: double-sided Braille image dataset and algorithm evaluation for Braille dots detection. In Proceedings of the 2018 the 2nd International Conference on Video and Image Processing. 65–69.

[7] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. 2017. Focal loss for dense object detection. In Proceedings of the IEEE international conference on computer vision. 2980–2988

[8] Ilya G Ovodov. 2021. Optical Braille Recognition Using Object Detection Neural Network. In Proceedings of the IEEE/CVF International Conference on Computer Vision. 1741–1748.

[9] Taekeun Park, Jungil Jung, and Jinsoo Cho. 2016. A method for automatically translating print books into electronic Braille books. Science China Information Sciences 59, 7 (2016), 1–14.

[10] Dian Utami and Annisa Salsabilla. 2021. Constraints of Teacher in The Process of Learning Geography Blind Students in SLB Bandar Lampung. In ICOPE 2020: Proceedings of the 2nd International Conference on Progressive Education, ICOPE 2020, 16-17 October 2020, Universitas Lampung, Bandar Lampung, Indonesia. European Alliance for Innovation, 60.

[11] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. Advances in neural information processing

Authorized licensed use limited to: University of New South Wales. Downloaded on December 02,2025 at 14:30:55 UTC from IEEE Xplore. Restrictions apply.

systems 30 (2017).

[12] Ju-Xiao Zhang, Hai-Feng Chen, Bing Chen, Bei-Qin Chen, Jing-Hua Zhong, and Xiao-Qin Zeng. 2022. Design and Implementation of Chinese Common Braille Translation System Integrating Braille Word Segmentation and Concatenation Rules. Computational Intelligence and Neuroscience 2022 (2022).

[13] Shanjun Zhang and Kazuyoshi Yoshino. 2007. A braille recognition system by the mobile phone with embedded camera. In Second International Conference on Innovative Computing, Informatio and Control (ICICIC 2007). IEEE, 223–223.

Authorized licensed use limited to: University of New South Wales. Downloaded on December 02,2025 at 14:30:55 UTC from IEEE Xplore. Restrictions apply.